JANLITA

Janlita Technology

Microphone Department



Customer Name:

Customer Model:

Customer P/N:

Janlita P/N:CMO-627BA283C-104P

Versions: 1.0

JANLITA	CUSTOMER APPROVAL
	

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Specification history

Date	Revision level	Description of changes
2025-11-12	1.0	Initial release

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1 Scope

This specification shall be applied for back-electrode electret condenser microphone (Back-Electrode ECM).

2 Microphone model number

CMO-627BA283C-104P

3 Electrical characteristics

Test Temperature: 15℃ ~ 35℃

Test Room Humidity: 40% ~ 85 %R.H.

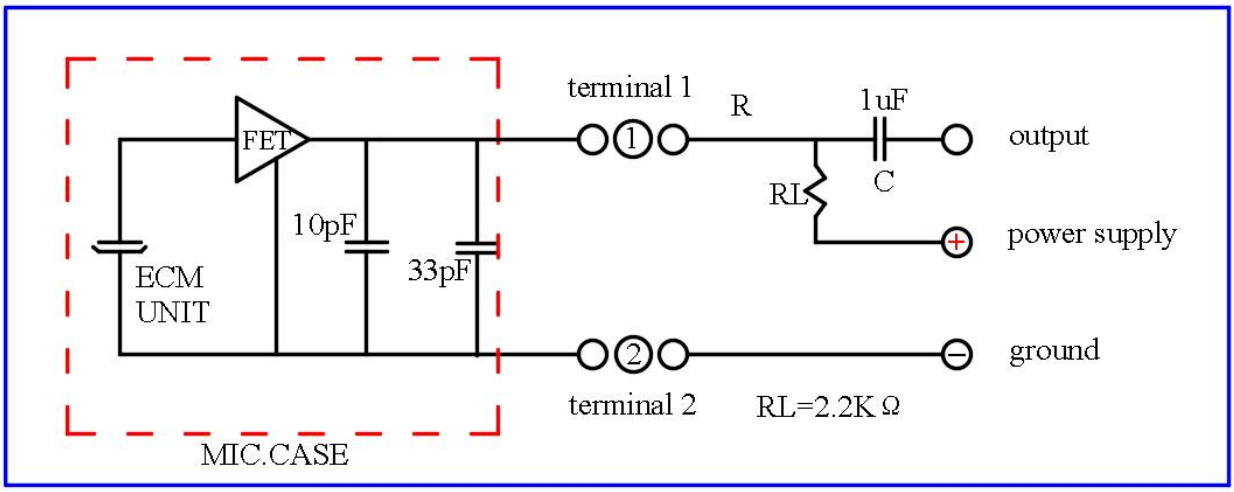
NO	Parameter	Condition	Limits			Unit
			Min	Nom	Max	
1	Directivity	Omnidirectional				
2	Sensitivity	1kHz(0dB=1V/Pa)	-31	-28	-25	dB
3	Output impedance	f=1kHz			2.2	kΩ
4	Current Consumption	VCC=2.0V, RL =2.2kΩ			0.5	mA
5	Operating Voltage		2.0		5.0	V
6	Total Harmonic Distortion	f=1kHz, Pin=94dB			1	%
7	S/N Ratio	f=1kHz, Pin=1Pa (A-Weighted curve)		65		dB
8	Acoustical Overload Point	＜10% THD @1kHz			115	dB SPL

Temperature conditions

Storage temperature: -10℃ ~ 60℃, With 40% ~ 85% R.H. humidity.

Operating temperature: -10℃ ~ 60℃, With 40% ~ 85% R.H. humidity.

4 Measurement circuit



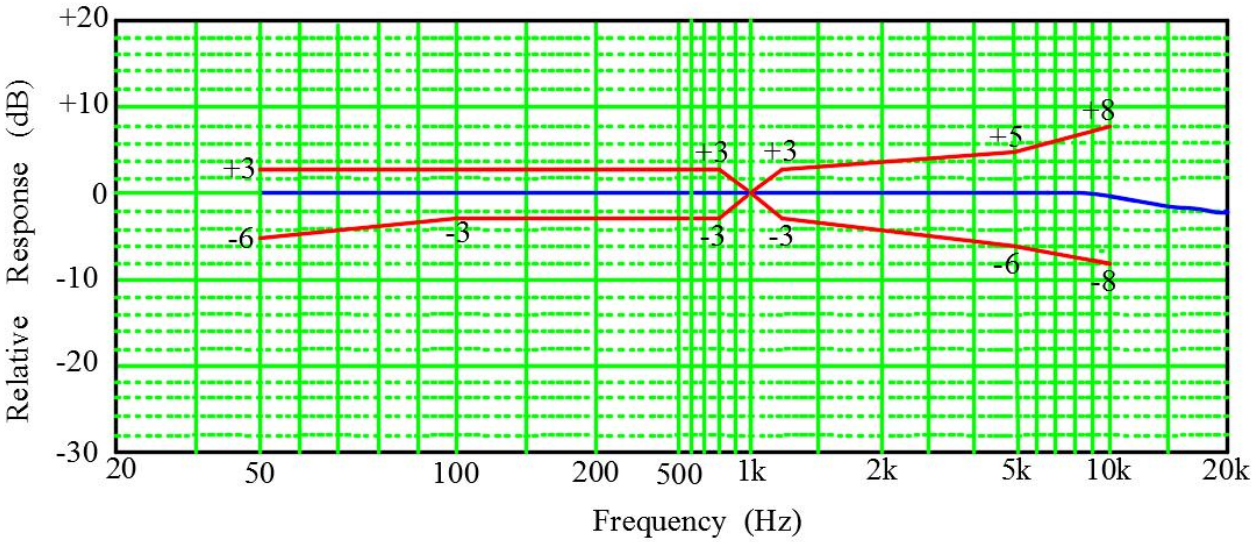
5 Typical frequency response curve

Test temperature: 15℃ ~ 35 ℃

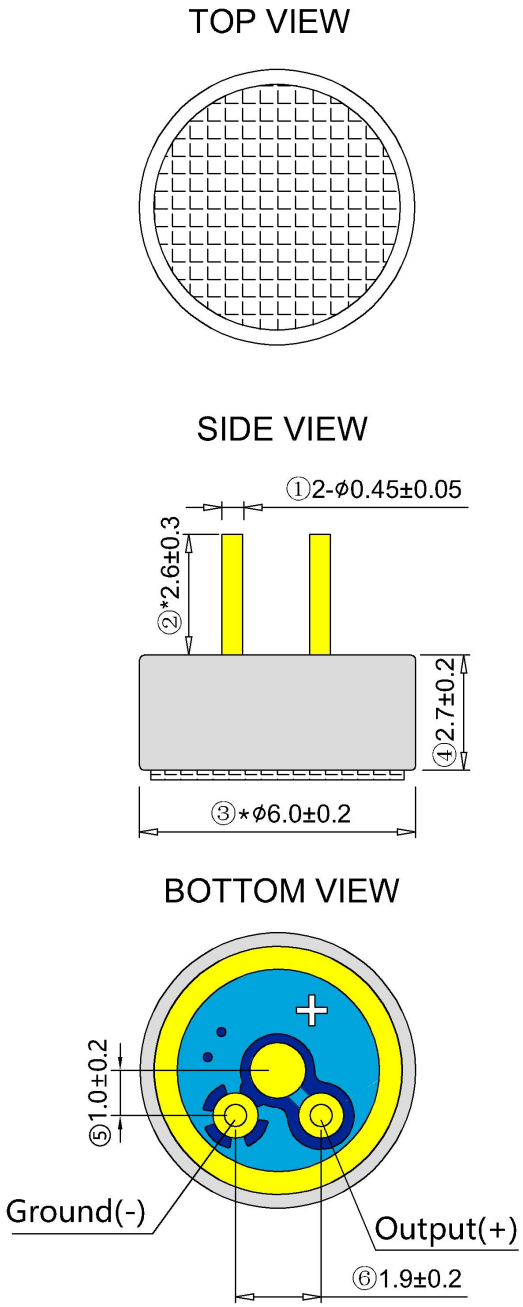
Acoustic stimulus: 1Pa (94dB SPL at 1kHz) at 50cm from the loud-speaker.

The loud-speaker must be calibrated to make a flat frequency response curve input signal.

Placement Location: The frequency response of microphone unit measured at 50cm from the loud-speaker.



6 Microphone outline dimensions



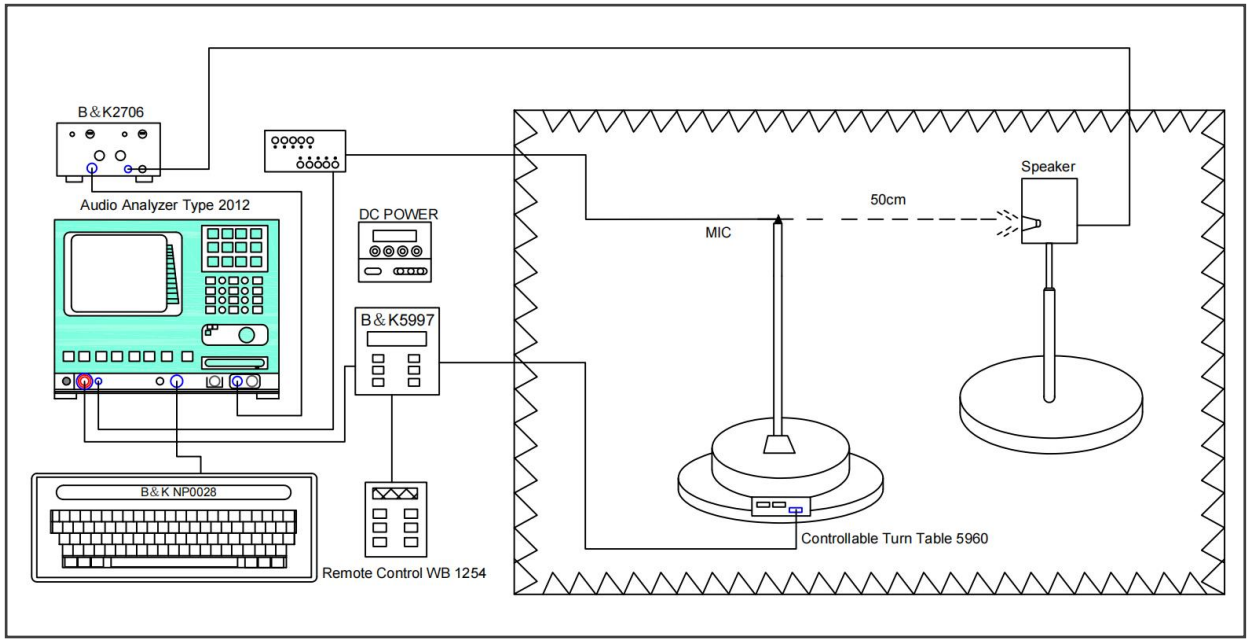
Note:
All dimensions are in millimeter(mm).

Tolerance：+/-0.1mm unless otherwise specified.

The dimensions marked with * are the key controlled dimensions.

7 Measurement system

The specification for the far-field measurement system conforms to IEC60268-4.



8 Reliability test

Sensitivity difference shall be within ±3dB (at 1kHz) between before and after the experiment, Wait for 2 hours to test after completing experiment.

Ambient temperature: 15℃ ~ 35℃.

8.1 Drop Test

Height:1m

Drop Face:2cm hardwood board

Duration: 3 times in each of three directions in state of packing

8.2 Vibration Test

Frequency range: 10~55Hz

Accelerate: 3mm amplitude

Duration per axis: 30minutes vibration at three axes.

8.3 High Temperature Test

Temperature: 80℃

Duration: 96 hours

8.4 Low Temperature Test

Temperature: -40℃

Duration: 96 hours

8.5 Temperature Cycle Test

Low temperature: -25±2℃ X 30min ->

Normal temperature: 20℃±2℃ X 10min ->

High temperature: 70℃±2℃ X 30min ->

Normal temperature: 20℃±2℃ X 10min)

No. of repetitions: 5

8.6 Humidity Test

Temperature: 70℃

Humidity: 90~95% R.H.

Duration: 96 hours

8.7 Soldering Heat Shock Test

350±10℃ with in 3 second Max for each terminal

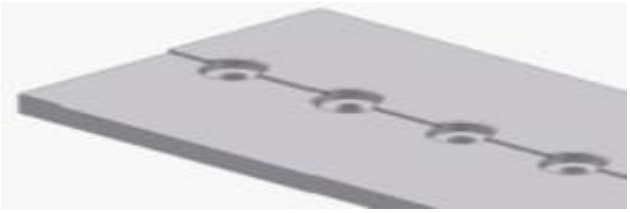
9 Regarding the soldering operation

9.1 Soldering Condition

Every ECM contains a FET with microphone body. This FET is easy to damageable from excessive heat and electrical shock. Proper attention for the soldering work is required same as followings.

Recommend to use high temperature soldering iron and apply 350±10℃ temperature range, Soldering should be accomplished within 1-3 seconds at each terminal so as not to be overheated. Do not make a cavity at the surface on the PCB and wiring board (Opened cavity will influence to the sensitivity of ECM MIC). Optimal design for heat sink pad is same as below.

9.2 Soldering fixture

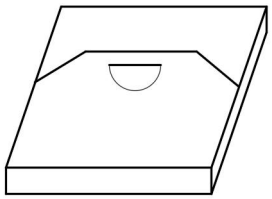
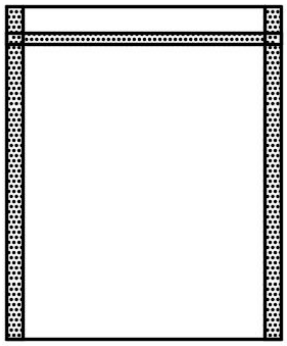
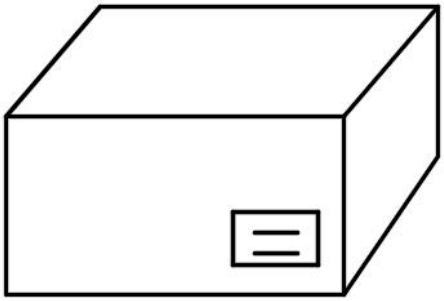


ECM eject hole.

Solder pad jig hole size: microphone diameter + 0.05㎜.

Solder pad jig hole height: microphone height –0.2㎜.

10 The graphic of packing method as follows

	Drawing(Unit: mm)	Packing (PCS)	Qty (PCS)	Material
Packing		1*100	100	Paper
Middle bag		10*100	1000	Anti static bag
Outer Box		10*1000	10000	Paper

Special note:

The above packaging method is universal packaging. If the actual shipment order quantity is less than the full box quantity or there is a slight surplus, the cardboard box size and product quantity will be adjusted. we won't notice again, if you have any questions, please contact our marketing department.